

Data Analytics using Python

Week 2 Question Bank Answer

1 What is probability? What is range of probability?

Ans.1) Probability is a numerical measure of the likelihood that an event will occur. It is a number between 0 and 1, where 0 represents an impossible event and 1 represents a certain event. The sum of the probabilities of all mutually exclusive and collectively exhaustive events is 1

For example, if you roll a fair six-sided die, the probability of rolling a 3 is $\frac{1}{6}$, because there is one favorable outcome (rolling a 3) out of six possible outcomes (rolling a 1, 2, 3, 4, 5, or 6). The probability of rolling an even number (2, 4, or 6) is $\frac{1}{2}$, because there are three favorable outcomes out of six possible outcomes.

The range of probability is from 0 to 1, inclusively. This means that the lowest possible probability is 0, which represents an impossible event, and the highest possible probability is 1, which represents a certain event. Any probability between 0 and 1 represents a possible, but not certain, event.

There are several methods of assigning probability, including the classical method, the relative frequency method, and the subjective method. The classical method is based on the number of ways an event can occur divided by the total number of possible outcomes. The relative frequency method is based on the number of times an event has occurred divided by the number of trials. The subjective method is based on a person's intuition or reasoning

In summary, probability is a numerical measure of the likelihood of an event, and it ranges from 0 to 1. It can be assigned using various methods, including the classical, relative frequency, and subjective methods.

2 What are Methods of Assigning Probabilities?

Ans.2) The methods of assigning probabilities include the classical method, relative frequency method, and subjective probability method¹.

1. Classical Method:

- This method is based on rules and laws.
- It involves determining the number of outcomes leading to an event divided by the total number of possible outcomes.
- Each outcome is assumed to be equally likely and is determined a priori before the experiment.
- It is applicable to games of chance and is objective, meaning everyone using the method assigns the same probability to an event¹.

2. Relative Frequency Method:

- This method is based on historical data.
- Probabilities are computed after performing the experiment.
- It involves dividing the number of times an event occurred by the total number of trials.
- Like the classical method, it is objective, with everyone correctly using the method assigning the same probability to an event¹.

3. Subjective Probability Method:

- This method is based on personal intuition or reasoning.
- It is subjective, meaning different individuals may assign different numeric probabilities to the same event based on their beliefs.
- It is useful for unique experiments like new product introductions, stock offerings, site selections, and sporting events where historical data may not be available¹.

These methods provide different approaches to assigning probabilities, catering to various scenarios and types of experiments.

3 Define the following: Probability - Terminology

- i) Experiment
- ii) Event
- iii) Elementary Events
- iv) Sample Space
- v) Unions and Intersections
- vi) Mutually Exclusive Events
- vii) Independent Events
- viii) Collectively Exhaustive Events
- ix) Complementary Events

Ans.3) Probability - Terminology Definitions

i) Experiment:

- An experiment is a process that produces outcomes where there is more than one possible outcome, and only one outcome occurs per trial.

ii) Event:

- An event is an outcome of an experiment, which can be either an elementary event (indivisible) or an aggregate of elementary events.

iii) Elementary Events:

- Elementary events are outcomes that cannot be further decomposed or broken down into other events.

iv) Sample Space:

- The sample space is the set of all elementary events for an experiment, providing a comprehensive description of all possible outcomes.

v) Unions and Intersections:

- **Unions:** The union of two sets contains an instance of each element from both sets.
- **Intersections:** The intersection of two sets contains only those elements common to both sets.

vi) Mutually Exclusive Events:

- Mutually exclusive events are events with no common outcomes, where the occurrence of one event precludes the occurrence of the other event.

vii) Independent Events:

- Independent events are events where the occurrence of one event does not affect the occurrence or non-occurrence of the other event.

viii) Collectively Exhaustive Events:

- Collectively exhaustive events contain all elementary events for an experiment, ensuring that every possible outcome is covered.

ix) Complementary Events:

- Complementary events consist of all elementary events not in a specific event 'A,' representing the outcomes that are not part of event 'A.'

These definitions provide a comprehensive understanding of the key terminology related to probability, helping to clarify the fundamental concepts involved in probability theory.

4 What do you mean by Sample Space? what are different Methods for describing a sample space?

Ans.4) Sample Space and Methods for Describing a Sample Space

Sample Space:

- The sample space is the set of all possible outcomes of an experiment.
- It provides a comprehensive description of all possible outcomes for an experiment.
- Sample spaces are crucial in probability theory for analyzing events and assigning probabilities.

Methods for Describing a Sample Space:

1. Roster or Listing:

- Involves listing all possible outcomes explicitly.
- Suitable for experiments with a limited number of outcomes.

2. Tree Diagram:

- Represents outcomes in a branching structure, showing the different paths and outcomes.
- Useful for visualizing complex experiments with multiple stages.

3. Set Builder Notation:

- Describes the sample space using set notation.
- Concise representation, especially for experiments with numerous outcomes.

4. Venn Diagram:

- Illustrates relationships between sets and outcomes.
- Useful for visualizing intersections and unions of events.

These methods provide different ways to describe a sample space, catering to various types of experiments and helping in the analysis of probabilities and events.

5 Problem on Sampling from a Population with Replacement

- A tray contains 1,000 individual tax returns. If 3 returns are randomly selected with replacement from the tray, how many possible samples are there?
- $(N)n = (1,000)3 = 1,000,000,000$

Ans.5) The number of possible samples when 3 returns are randomly selected with replacement from a tray containing 1,000 individual tax returns is calculated using the formula $(N)n$, where N is the number of items in the population and n is the number of items in the sample.

In this case, $N = 1,000$ (individual tax returns) and $n = 3$ (returns selected).

Therefore, the number of possible samples is $(1,000)3 = 1,000,000,000$.

Hence, there are 1,000,000,000 different ways to select 3 tax returns with replacement from a tray containing 1,000 individual tax returns.

6 A tray contains 1,000 individual tax returns. If 3 returns are randomly selected without replacement from the tray, how many possible samples are there?

Ans.6) The problem asks for the number of possible samples when 3 returns are randomly selected without replacement from a tray containing 1,000 individual tax returns.

The number of possible samples is calculated using the combination formula, which is a type of counting method. The formula is:

$$n! / [k!(n-k)!]$$

where n is the total number of items in the population, and k is the number of items in the sample.

In this case, n = 1,000 (individual tax returns) and k = 3 (returns selected). Therefore, the number of possible samples is:

$$1,000! / [3!(1,000-3)!] = 166,167,000$$

Hence, there are 166,167,000 different ways to select 3 tax returns without replacement from a tray containing 1,000 individual tax returns.

7 Give four types of probability.: Marginal, Joint, Union, Conditional.

Ans.7) The four types of probability are:

1. **Marginal Probability:** This is the probability of an event occurring without considering any other events. It is calculated by summing the joint probabilities of the event with all possible outcomes of the other events.

For example, if you roll a fair six-sided die, the marginal probability of rolling a 3 is $1/6$.

2. **Joint Probability:** This is the probability of two or more events occurring together. It is calculated by multiplying the individual probabilities of each event.

For example, if you draw two cards from a standard deck of 52 cards without replacement, the joint probability of drawing a heart and a diamond is 0. The events of drawing a heart and a diamond are mutually exclusive, so their intersection is empty.

3. **Union Probability:** This is the probability of an event occurring in at least one of two or more sets of outcomes. It is calculated by adding the probabilities of each set and subtracting the joint probabilities of the sets.

For example, if you draw two cards from a standard deck of 52 cards without replacement, the union probability of drawing a heart or a diamond is $26/52$ or $1/2$.

4. **Conditional Probability:** This is the probability of an event occurring given that another event has occurred. It is calculated by dividing the joint probability of the two events by the marginal probability of the second event.

For example, if you draw two cards from a standard deck of 52 cards without replacement, the conditional probability of drawing a heart given that the first card was a heart is $12/51$.

These probabilities are used to analyze and understand the relationships between different events in a sample space.

8 Give the general law of addition.

Ans.8) The general law of addition in probability is used to calculate the probability of the union of two events, which is the probability of either event occurring. The general law of addition states that the probability of the union of two events A and B is equal to the sum of the probability of each event minus the probability of their intersection.

Mathematically, it is expressed as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Where:

- $P(A \cup B)$ is the probability of the union of A and B
- $P(A)$ and $P(B)$ are the probabilities of each event
- $P(A \cap B)$ is the probability of the intersection of A and B, which is the probability of both events occurring together.

The general law of addition is used when the events are not mutually exclusive, meaning that they can occur at the same time. If the events are mutually exclusive, then the probability of their intersection is zero, and the formula simplifies to:

$$P(A \cup B) = P(A) + P(B)$$

This is because the probability of the union of two mutually exclusive events is the sum of their probabilities.

Here is an example of the general law of addition:

Suppose you roll two dice, and you want to find the probability of rolling a 3 or a 4.

The sample space is:

(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)
(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)
(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)
(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)
(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

The event A is rolling a 3, and the event B is rolling a 4.

The union of A and B is rolling a 3 or a 4.

The intersection of A and B is rolling a 3 and a 4 at the same time, which is impossible. Therefore, the probability of the intersection is zero.

The probability of rolling a 3 is $2/36$, and the probability of rolling a 4 is $3/36$.

Therefore, the probability of rolling a 3 or a 4 is:

$$P(A \cup B) = P(A) + P(B) = 2/36 + 3/36 = 5/36$$

This is the probability of rolling a 3 or a 4 using the general law of addition.

- 9 **Problem:** If one of the survey respondents was randomly selected and asked what office design changes would increase worker productivity, what is the probability that this person would select reducing noise or more storage space?

Solution

- *Let N represent the event "reducing noise."*
- *Let S represent the event "more storage/ filing space."*
- *The probability of a person responding with N or S can be symbolized statistically as a union probability by using the law of addition.*

Ans.9) The problem is about calculating the probability of a survey respondent selecting "reducing noise" or "more storage space" as office design changes that would increase worker productivity.

The events are defined as:

- **N: Reducing noise**
- **S: More storage/filing space**

The probability of a person responding with N or S can be symbolized statistically as a union probability by using the law of addition:

$$P(N \text{ or } S) = P(N) + P(S) - P(N \text{ and } S)$$

The general law of addition states that the probability of the union of two events A and B is equal to the sum of the probability of each event minus the probability of their intersection.

The probability of each event can be obtained from the survey data:

- **$P(N) = 70\%$**
- **$P(S) = 67\%$**

The probability of both events occurring together, $P(N \text{ and } S)$, is not provided in the problem. However, if it is assumed that the events are mutually exclusive, meaning that they cannot occur at the same time, then the probability of their intersection is zero.

Therefore, the probability of a person responding with N or S can be calculated as:

$$\begin{aligned} P(N \text{ or } S) &= P(N) + P(S) - P(N \text{ and } S) \\ &= P(N) + P(S) - 0 \\ &= 70\% + 67\% \\ &= 137\% \end{aligned}$$

However, the probability of an event cannot be greater than 100%. This indicates that the assumption of mutual exclusivity is incorrect, and the events are not mutually exclusive.

Therefore, the probability of a person responding with N or S should be calculated as:

$$P(N \text{ or } S) = P(N) + P(S) - P(N \text{ and } S)$$

The value of $P(N \text{ and } S)$ needs to be obtained from the survey data or estimated based on the context of the problem. Once the value of $P(N \text{ and } S)$ is known, the probability of a person responding with N or S can be calculated accurately.

10 Give Law of Multiplication.

Ans.10) The Law of Multiplication in probability states that the probability of two independent events occurring together is the product of the probability of each event.

Mathematically, it can be expressed as:

$$P(A \text{ and } B) = P(A) * P(B)$$

where:

- $P(A \text{ and } B)$ is the probability of both events A and B occurring together.
- $P(A)$ is the probability of event A occurring.
- $P(B)$ is the probability of event B occurring.

This law is applicable when events A and B are independent, meaning the occurrence of one event does not affect the probability of the other event. It is a fundamental concept in probability theory used to calculate the joint probability of events in various scenarios.

For example:

Suppose we have two independent events A and B, such that $P(A) = 0.6$ and $P(B) = 0.3$. Then, the probability of both events A and B occurring together can be calculated as:

$$P(A \text{ and } B) = P(A) * P(B) = 0.6 * 0.3 = 0.18$$

Therefore, the probability of both events A and B occurring together is 0.18.

11 Law of Conditional Probability

- The conditional probability of X given Y is the joint probability of X and Y divided by the marginal probability of Y.

Ans.11) The Law of Conditional Probability is a fundamental concept in probability theory that describes the probability of an event A occurring given that another event B has occurred. It is expressed as the joint probability of A and B divided by the marginal probability of B, mathematically represented as:

$$P(A|B) = P(A \cap B) / P(B)$$

where:

- $P(A|B)$ is the conditional probability of A given B
- $P(A \cap B)$ is the joint probability of A and B
- $P(B)$ is the marginal probability of B

This law is used to calculate the probability of an event based on the occurrence of another event, which is useful in various applications, such as hypothesis testing and decision making.

Example:

Suppose we have two events A and B, where $P(A) = 0.4$, $P(B) = 0.6$, and $P(A \text{ and } B) = 0.2$. We want to find the conditional probability of A given B, denoted as $P(A|B)$.

Using the Law of Conditional Probability, we have:

$$P(A|B) = P(A \text{ and } B) / P(B)$$

Substituting the given values, we get:

$$P(A|B) = 0.2 / 0.6 = 1/3 \approx 0.333$$

Therefore, the conditional probability of A given B is approximately 0.333

12 What is distribution?

Ans.12) Distribution

- A distribution is a mathematical function that describes the probability of different outcomes in a random experiment.
- It provides a summary of the data and its characteristics, which can be used to compare and analyze empirical distributions.
- There are different types of distributions, including discrete and continuous distributions.
 - i) **Discrete distributions** - distinct values that can be counted
 - ii) **Continuous distributions** - a mass of values within a range.
- The characteristics of a distribution can be defined using a small number of numeric descriptors called 'parameters'.
- These parameters can include the mean, median, mode, variance, and standard deviation, among others.
- Distributions are important in statistics because they can serve as a basis for standardized comparison of empirical distributions, help estimate confidence intervals for inferential statistics, and form a basis for more advanced statistical methods.
- A probability distribution is used to describe the probability of different outcomes in a random experiment, such as the probability of selecting a female or a professional worker from a company's employees.
- The distribution can be represented using a probability mass function (PMF) for discrete distributions or a probability density function (PDF) for continuous distributions.
- **Example**
 - A company has 140 employees, of which 30 are supervisors. Eighty of the employees are married, and 20% of the married employees are supervisors.
 - If a company employee is randomly selected, what is the probability that the employee is married and is a supervisor?
 - The probability of an employee being married and a supervisor is calculated as the product of the probability of selecting a married employee and the probability of selecting a supervisor given that the employee is married.

In summary, a distribution is a mathematical function that describes the probability of different outcomes in a random experiment.

It is an important concept in statistics and can be used to analyze and compare empirical distributions.

13 The quiz scores for a particular student are given below:

22, 25, 20, 18, 12, 20, 24, 20, 20, 25, 24, 25, 18

Find the variance and standard deviation.

Ans.13) The question asks for the variance and standard deviation of the quiz scores for a particular student.

To calculate the variance, we need to find the mean of the scores first. The mean is calculated as the sum of the scores divided by the number of scores.

- i) Mean (μ) :** The mean of a set of data is the sum of all the values divided by the number of values. It is denoted by the symbol μ .

Mean (μ) = (sum of all data points) / (number of data points)

$$\mu = (22 + 25 + 20 + 18 + 12 + 20 + 24 + 20 + 20 + 25 + 24 + 25 + 18) / 13$$

$$\mu = 224 / 13$$

$$\mu = 17.23$$

- ii) Variance (σ^2) :** The variance of a set of data is the average of the squared differences between each value and the mean. It is denoted by the symbol σ^2 .

Variance (σ^2) = (sum of (data point - mean)²) / (number of data points - 1)

$$\sigma^2 = [(22 - 17.23)^2 + (25 - 17.23)^2 + (20 - 17.23)^2 + (18 - 17.23)^2 + (12 - 17.23)^2 + (20 - 17.23)^2 + (24 - 17.23)^2 + (20 - 17.23)^2 + (20 - 17.23)^2 + (25 - 17.23)^2 + (24 - 17.23)^2 + (25 - 17.23)^2 + (18 - 17.23)^2] / (13 - 1)$$

$$\sigma^2 = 225.18$$

- iii) Standard deviation (σ) :** The standard deviation of a set of data is the square root of the variance. It is denoted by the symbol σ .

Standard deviation (σ) = sqrt(variance)

$$\sigma = \text{sqrt}(\sigma^2)$$

$$\sigma = 15.01$$

Therefore, the variance of the quiz scores for the student is 225.18 and the standard deviation is 15.01.

14 Define covariance and correlation coefficient.

Ans.14) Covariance and correlation coefficient are measures of the relationship between two random variables.

Covariance is a measure of the linear relationship between two random variables, X and Y. It is defined as the expected value of the product of the deviations of the two variables from their respective means. Mathematically, it is represented as:

$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

where μ_x and μ_y are the means of **X** and **Y**, respectively.

The correlation coefficient is a standardized measure of the linear relationship between two random variables. It is defined as the covariance of the two variables divided by the product of their standard deviations. Mathematically, it is represented as:

$$\rho(X, Y) = \text{Cov}(X, Y) / (\sigma_x * \sigma_y)$$

where σ_x and σ_y are the standard deviations of **X** and **Y**, respectively.

The correlation coefficient ranges from **-1 to 1**,
where

- **-1 indicates a perfect negative linear relationship,**
- **1 indicates a perfect positive linear relationship,**
- **0 indicates no linear relationship.**

In summary, covariance is a measure of the linear relationship between two random variables, while the correlation coefficient is a standardized measure of the strength and direction of the linear relationship between two random variables.

- 15 The weight of football players is normally distributed with a mean of 200 pounds and a standard deviation of 25 pounds. What is the probability of a player weighing more than 241.25 pounds? Also calculate the probability of a player weighing less than 250 pounds.

Ans.15) Covariance is a measure of the linear relationship between two random variables, X and Y. It is defined as:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

where $E[X]$ and $E[Y]$ are the expected values of X and Y, respectively. The covariance can take on any value between negative infinity and positive infinity, and it can be positive, zero, or negative.

The correlation coefficient, on the other hand, is a standardized measure of the linear relationship between two random variables, X and Y. It is defined as:

$$\text{Corr}(X, Y) = \text{Cov}(X, Y) / (\text{StdDev}(X) \cdot \text{StdDev}(Y))$$

where $\text{StdDev}(X)$ and $\text{StdDev}(Y)$ are the standard deviations of X and Y, respectively. The correlation coefficient is always between -1 and 1, where -1 indicates a perfect negative linear relationship, 0 indicates no linear relationship, and 1 indicates a perfect positive linear relationship.

In the given question, the weight of football players is normally distributed with a mean of 200 pounds and a standard deviation of 25 pounds. To find the probability of a player weighing more than 241.25 pounds, we can use the standard normal distribution table or a calculator with a built-in function for the cumulative distribution function (CDF) of the normal distribution.

$$P(X > 241.25) = 1 - P(X \leq 241.25) = 1 - \text{CDF}(241.25, 200, 25)$$

To find the probability of a player weighing less than 250 pounds, we can use the CDF of the normal distribution:

$$P(X < 250) = \text{CDF}(250, 200, 25)$$

The CDF of the normal distribution is a function that gives the probability that a random variable X with a normal distribution and a mean of μ and a standard deviation of σ will be less than or equal to a given value x. It is defined as:

$$\text{CDF}(x, \mu, \sigma) = (1/2) + (1/2) \cdot \text{erf}[(x - \mu) / (\sigma \cdot \sqrt{2})]$$

where erf is the error function, which is a special function defined as:

$$\text{erf}(x) = (2/\sqrt{\pi}) \cdot \int_0^x e^{-t^2} dt$$

The error function can be computed using a calculator or a computer program.

16 Assuming a binomial experiment with $p = 0.5$ and a sample size of 100. What is the expected value of this distribution?

Ans.16) The expected value of a binomial distribution is given by the formula:

$$E(X) = np$$

where n is the number of trials, p is the probability of success, and X is the number of successes in the trials. In this case, $n = 100$ and $p = 0.5$, so:

$$E(X) = 100 * 0.5 = 50$$

Therefore, the expected value of this distribution is **50**.